Case Study 1 — Employee Salaries

A company records the annual salaries (in ₹ lakhs) of 9 employees:

4, 5, 5, 6, 7, 8, 8, 9, 30

1. Find the **mean, median, and mode**.
2. Find the **range**.
3. Calculate the **variance and standard deviation**.
4. Identify the outlier.
5. Which measure of central tendency best represents the typical salary? Explain why

Case Study 2 — Website Response Time

A data scientist records the response time (in milliseconds) of a website for 10 requests:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

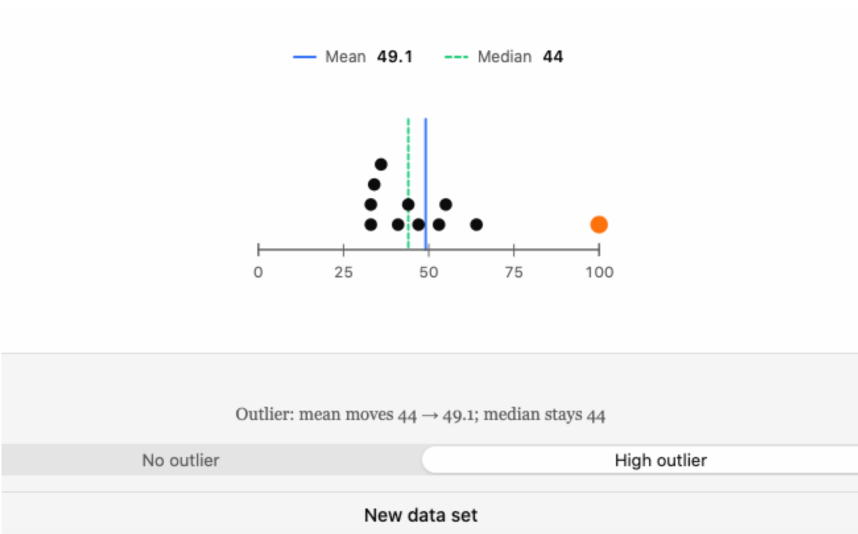
1. Calculate the mean and median.
2. Find the range.
3. Calculate the variance and standard deviation.
4. Determine whether the data is likely to be **right-skewed or left-skewed**.
5. Explain why the median may be more useful than the mean.

Case Study 3 — Student Performance

The marks of 12 students in a Data Science test are:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

1. Find Q1, Q2 and Q3.
2. Calculate the **IQR**.
3. Use the **1.5 × IQR rule** to identify any outliers.
4. Find the mean and median.
5. Explain how the outlier affects the mean.



Case Study 4 — E-Commerce Orders

An online store records the number of orders received over 10 days:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

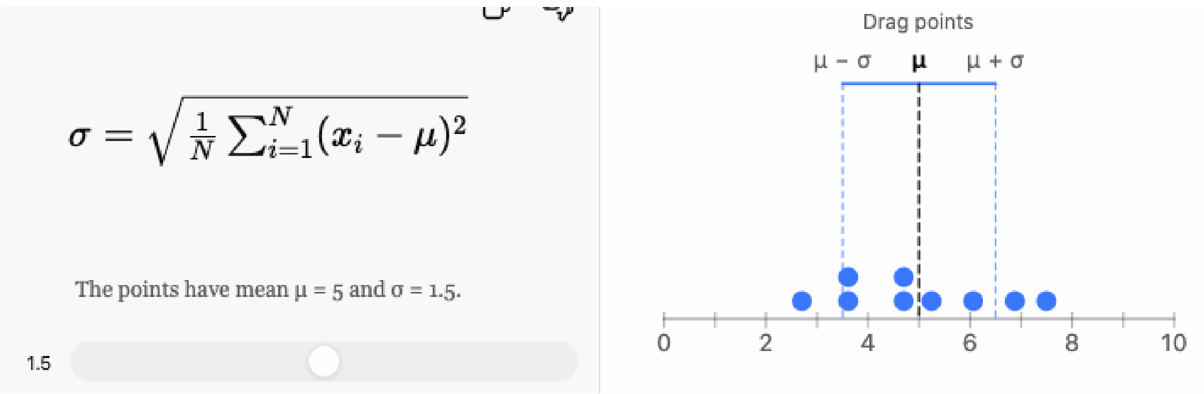
1. Calculate the mean, median, and mode.
2. Find the variance and standard deviation.
3. Calculate the coefficient of variation if the mean and standard deviation are known.
4. What does the standard deviation tell the company?
5. Would you describe the order data as highly or moderately variable?

Case Study 5 — Comparing Two Data Science Models

Two machine-learning models have the following prediction errors:

Model A: 2, 3, 3, 4, 4, 5, 5, 6
Model B: 1, 1, 2, 4, 5, 7, 8, 9

1. Calculate the mean error for both models.
2. Calculate the variance for both models.
3. Calculate the standard deviation for both.
4. Which model has more consistent predictions?
5. Can the model with the lower mean error necessarily be considered better? Explain.

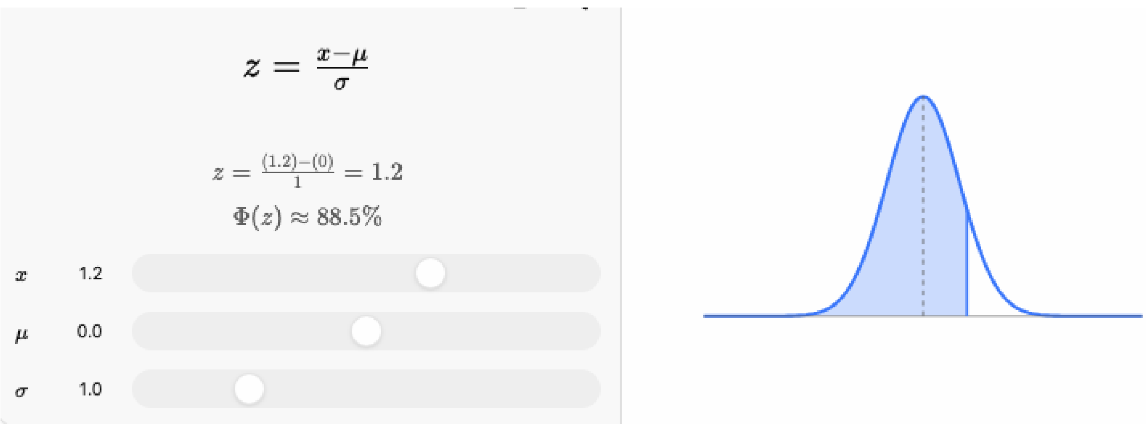


Case Study 6 — Detecting an Unusual Customer

A shopping website finds that the average amount spent by customers is **₹2,000**, with a standard deviation of **₹400**.

A particular customer spends **₹3,200**.

1. Calculate the customer's **z-score**.
2. Is the customer's spending unusually high compared with the average?
3. What would a negative z-score mean in this context?
4. If another customer has a z-score of −1.5, what does it tell you?



Case Study 7 — Delivery Time Analysis

A delivery company records delivery times (in minutes):

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

1. Calculate the mean and median.
2. Find Q1 and Q3.
3. Calculate the IQR.
4. Check for outliers using the 1.5 × IQR rule.
5. Explain whether the mean or median gives a better description of the typical delivery time.

Case Study 8 — Two Cities' Temperatures

The average daily temperatures (°C) for 7 days are:

City A: 28, 29, 30, 30, 31, 30, 32
City B: 20, 25, 30, 35, 30, 25, 45

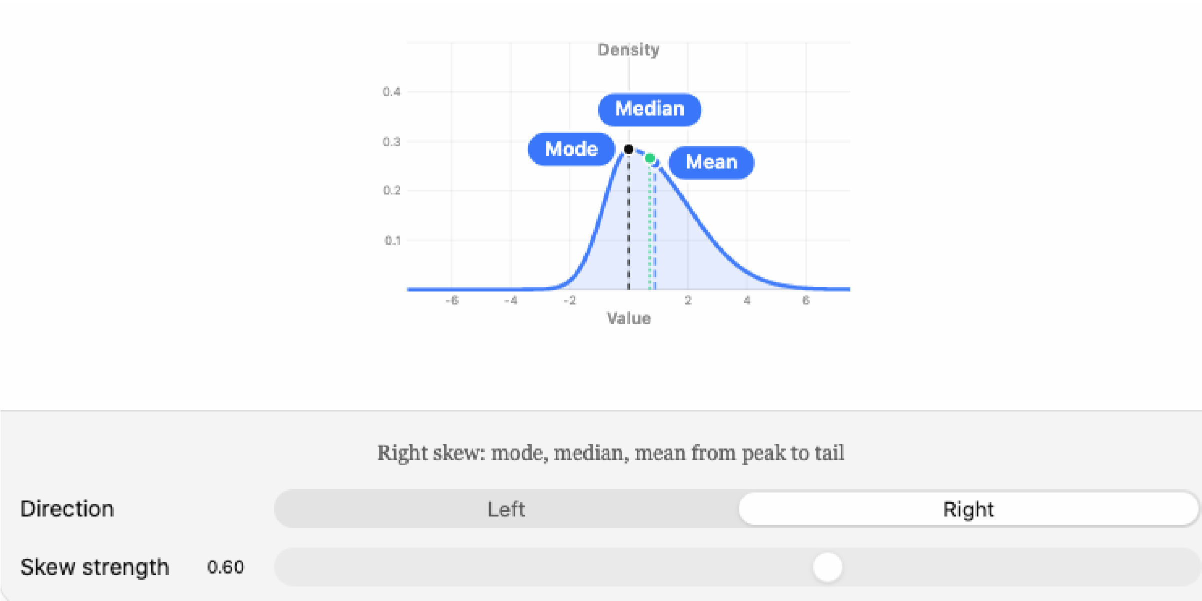
1. Calculate the mean temperature for each city.
2. Calculate the variance for each city.
3. Which city has greater temperature variability?
4. Why can two datasets have the same mean but different standard deviations?
5. Which city has more predictable temperatures?

Case Study 9 — Data Center CPU Usage

A data scientist records CPU usage percentages:

40, 42, 45, 45, 46, 48, 50, 52, 55, 90

1. Calculate mean, median, and mode.
2. Calculate the range.
3. Find Q1, Q3 and IQR.
4. Identify possible outliers.
5. Determine the direction of skewness.



Case Study 10 — Comparing Two Datasets

Two data analysts receive the following datasets:

Dataset A: 10, 20, 30, 40, 50
Dataset B: 28, 29, 30, 31, 32

1. Calculate the mean of both datasets.
2. Calculate the variance of both.
3. Calculate the standard deviation of both.
4. Which dataset has greater spread?
5. If both datasets represent model errors, which model appears more consistent?
6. Explain why **mean alone is not enough** to compare datasets.